

R Programming

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Tutorial One

February 24, 2015

1 What is R ?

“R is an integrated suite of software facilities for data *manipulation, calculation and graphical display*.

It includes

- An effective data handling and storage facility
- A suite of operators for calculations on arrays, in particular matrices
- Large, coherent, integrated collection of intermediate tools for data analysis
- Graphical facilities for data analysis and display either on-screen or on hardcopy
- A well-developed, simple and effective programming language which includes conditionals, loops, user-defined recursive functions and input and output facilities.“
-<http://www.r-project.org/>

R is a dialect of the **S** language developed by **John Chambers** at **Bell Labs** in **1976**

2 Why R?

- Available for almost all operating systems
- Lingua franca of the statistical community
- State-of-the-art graphics capabilities
- General purpose programming language
- Contains advanced statistical routines
- Integrates with other languages/software
- Good online documentation & vibrant user community
- Free

3 Installation

Visit <http://cran.us.r-project.org> for installing R

Visit <http://www.rstudio.com/> for installing RStudio which is a user-friendly IDE for R

Note: Install R first and then RStudio

4 RStudio Interface

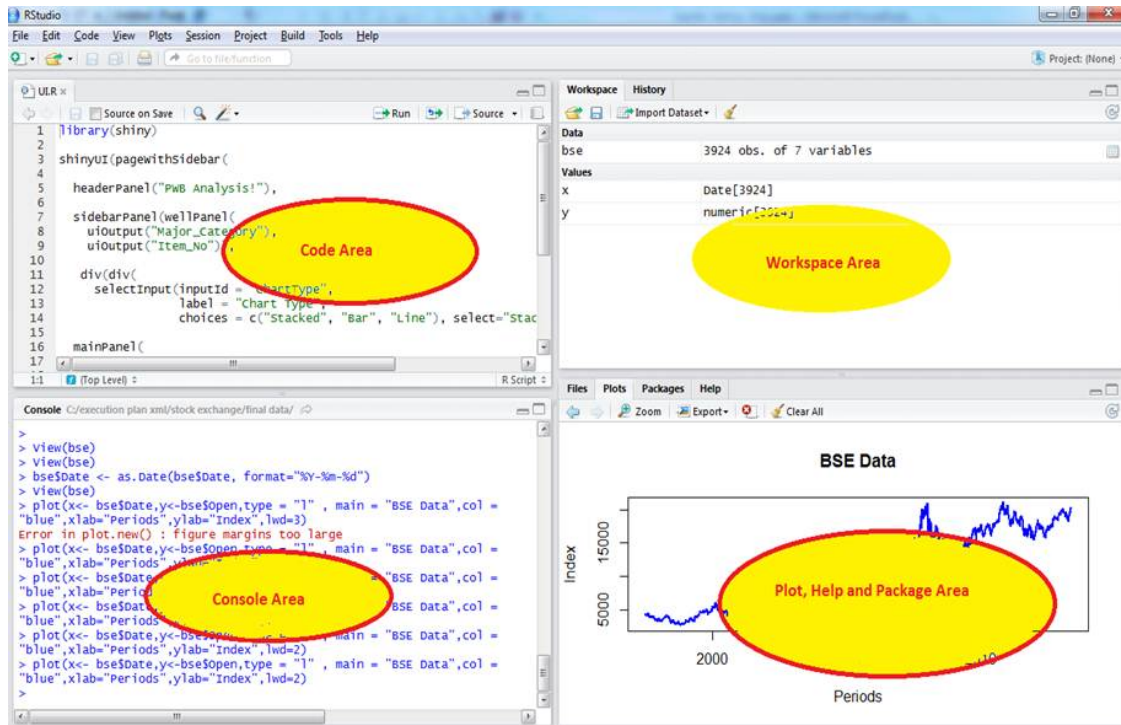


Figure 1: RStudio Interface

The interface of RStudio is divided into 4 quadrants:

- **R Script** - (upper left) All R codes are generally written here. You can save & edit these scripts
- **R Console**-(lower left) You can write R commands here for small tasks. Commands entered here are immediately executed. All R codes are submitted to the R Console for execution
- **R Environment/History**-(upper right) *The Environment* tab shows the objects available to the console. These are subdivided into data, values (non-dataframe, non-function objects). As commands are entered in the console, they appear in the *History* tab. These histories can be saved and loaded, there is a search feature to locate previous commands, and individual lines or sections can be transferred back to the console and functions

- **Files/Plots/Packages/Help**-(lower right) The *Files* tab provides a simple file manager. It can be navigated in familiar ways and used to open, move, rename, and delete files. The *Help* tab is where RStudio displays R help files. These can be searched and navigated in the Help tab. Plots created in the console are displayed in the *Plots* tab. Much of the functionality of R is located in packages, many of which can be obtained from a central clearing house called CRAN (Comprehensive R Archive Network). The *Packages* tab facilitates installing and loading packages. It will also allow you to search for packages that have been updated since you installed them.

5 First R Session notes

- R is a interpreted, command based language
- R is case sensitive
- Statements need not end with a semicolon. However if multiple statements are written in a line, semicolon is required.
- Comments begin with # symbol
- Results of calculations can be stored in objects using the assignment operators:
An arrow (<-) by a smaller than character and a hyphen without a space
The equal character (=)
- Object names cannot contain ‘strange’ symbols like !, +, -, # A dot (.) and an underscore (_) are allowed, also a name starting with a dot.
- Object names can contain a number but cannot start with a number
- Explicit print statements are not necessary
- Script files are saved with .R extension
- Quit R using q() command or File menu->Exit

6 R Help

Plethora of resources are available to help us learn about R. This include several facilities within R and the Web

6.1 Internal Help

Function	Action
<code>help.start()</code>	Getting help
<code>help("foo")</code> or <code>?foo</code>	Help on function foo (the quotation marks are optional)
<code>help.search("foo")</code> or <code>??foo</code>	Search the help system for instances of the string foo
<code>help(package=foo)</code>	Learn about entire package
<code>example("foo")</code>	Examples of function foo (the quotation marks are optional)
<code>RSiteSearch("foo")</code>	Search for the string foo in online help manuals and archived mailing lists
<code>apropos("foo", mode="function")</code>	List all available functions with foo in their name
<code>data()</code>	List all available example datasets contained in currently loaded packages

6.2 Web Help

- Manuals available from <http://www.r-project.org/>
- Search engines listed on R home page. Click Search “sos” package in R
- Rseek search engine: <http://www.rseek.org/>
- <http://stackoverflow.com/>
- <http://r.789695.n4.nabble.com/>

7 R Workspace

Objects (vectors, matrices, functions, data frames, or lists). that you create during an R session are held in **memory**, the collection of objects that you currently have is called the workspace. This workspace is not saved on disk unless you tell R to do so.

Function	Action
<code>getwd()</code>	List the current working directory
<code>setwd("mydirectory")</code>	Change the current working directory to mydirectory
<code>help(ls())</code>	List the objects in the current workspace
<code>rm(objectlist)</code>	Remove (delete) one or more objects

8 R Packages

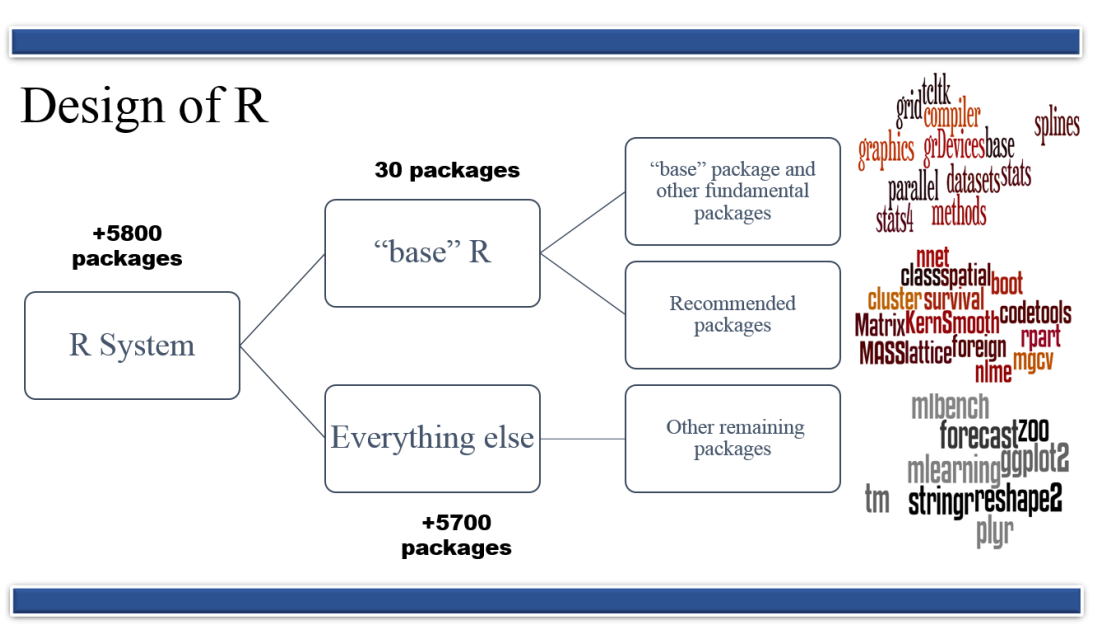


Figure 2: Design of R

Functionality of R is divided into a number of **packages**. Packages are collections of R functions, data, and compiled code in a well-defined format. The directory where packages are stored on your computer is called the **library**.

Function	Action
library()	Shows packages stored in
search()	Which packages are loaded
install.packages(packagename,dependencies=TRUE)	Loads required package as well as dependent packages
library(packagename)/require(packagename)	Load package
help(package="package_name")	Help about required package
detach(package:package_name,unload=TRUE)	Unload package
ls("package:package_name", all = TRUE)	Shows all functions and datasets in package